

Manish Khadka

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Professional Summary

AWS Certified DevOps Engineer (Professional) with 7+ years designing, automating, and optimising cloud-native infrastructure across AWS, GCP, and Azure. Specialise in Kubernetes at scale, Terraform-based IaC, CI/CD architecture, and DevSecOps. Known for delivering measurable cost reductions, building compliance-ready infrastructure, and solving non-standard problems — from ML inference on GPU clusters to air-gapped government deployments and blockchain node operations.

Key Achievements

- Reduced AWS monthly spend from \$24K to \$12K (**50%**) through right-sizing, spot instances, and OpenStack consolidation; replicated similar savings at a second company via reserved instances and automated scheduling.
- Led **SOC2 and ISO 27001** certification initiatives at two companies, hardening infrastructure controls and passing audits.
- Promoted from DevOps Engineer to **Team Lead**; built and mentored a 6-person team on IaC, GitOps, and cloud-native practices.
- Engineered **GPU sharing** across Kubernetes pods for real-time ML video analytics — a non-standard challenge that maximised inference utilisation.

Technical Skills

Cloud: AWS (primary), GCP, Azure, DigitalOcean, Oracle Cloud

Containers: Kubernetes (EKS, self-hosted), ECS, Docker, Containerd

CI/CD & GitOps: ArgoCD, GitLab CI, GitHub Actions, Jenkins, CircleCI, Bitbucket Pipelines

IaC & Config: Terraform, OpenTofu, Terragrunt, Ansible, CloudFormation, Packer

Observability: Prometheus, Grafana (Loki, Tempo, Alloy), ELK Stack, Datadog, PagerDuty

Security: Trivy, Snyk, Checkov, Kyverno, HashiCorp Vault, AWS Private CA, AWS Inspector

Networking: Kong API Gateway, Istio, Cilium, Nginx, Cloudflare (incl. Zero Trust), Route 53

Databases: PostgreSQL, MySQL, MongoDB, Redis, Elasticsearch, TimescaleDB, Milvus

Other: Helm, KEDA, HPA, External Secrets Operator, Snowflake, AWS Glue, Kinesis Firehose, MinIO

Practices: Site Reliability Engineering, Platform Engineering, GitOps, Zero Trust Architecture, Shift-Left Security, Chaos Engineering, Infrastructure as Code, Disaster Recovery, FinOps

Professional Experience

Zenledger

Senior DevOps Engineer

Remote

Feb 2025 – Present

- Cut monthly infrastructure costs by migrating edge security from Cloudflare to AWS WAF; built Python/Bash automation (Terraform-provisioned) to shut down QA/Dev environments outside business hours, eliminating idle spend.
- Designed and deployed a **High Availability TimescaleDB** cluster; orchestrated zero-downtime major-version migration of Amazon RDS from PostgreSQL 14 to 15.
- Replaced legacy VPN with **Cloudflare Zero Trust + Tailscale**, implementing identity-aware, least-privilege access across all environments.
- Led **SOC2 and ISO 27001** certification to completion; established VPC Peering across Prod/Stage/QA/Dev; managed rolling EKS and EC2 fleet upgrades with zero service disruption across **8+ clusters**.
- Designed production observability stack (Prometheus, Grafana, Grafana Alloy, PagerDuty) covering **100% of services**; established alerting thresholds aligned with SLA targets.

CloudHero

Senior DevOps Engineer

Remote

Nov 2023 – Feb 2025 (Contract)

- Migrated client services from on-premises to AWS, GCP, Azure, and DigitalOcean (ECS/EKS) with near-zero downtime; deployed self-hosted Kubernetes on GCP with Ansible and auto-scaling across **20+ clusters**.
- Authored a centralised **Helm chart blueprint** adopted company-wide; deployed Kyverno for admission control and policy enforcement across all client clusters.
- Migrated thousands of ingress resources to **Kong API Gateway**; implemented Istio service mesh with canary deployments for progressive rollouts.
- Eliminated static AWS secrets via **AWS Private CA** for IAM key rotation; integrated Trivy and Snyk for vulnerability scanning across all CI/CD pipelines.
- Deployed MinIO for S3-compatible storage and NextCloud for collaboration; built end-to-end monitoring with Prometheus, Grafana, Loki, and ELK Stack.

Innovatetech Pvt. Ltd

DevOps Engineer → Team Lead

Kathmandu, Nepal

Jul 2021 – Feb 2024

- Promoted to Team Lead; led and mentored **6 DevOps engineers** on IaC, GitOps, and cloud-native practices.
- Reduced cloud spend by **50%** across multi-account AWS (Control Tower, **10+ accounts**, VPC peering to MongoDB Atlas) through spot/reserved instances, right-sizing, and automated scheduling. Open-sourced a Terraform module for MongoDB Atlas provisioning.
- Built centralised GitLab CI pipelines with shared Helm charts and **shift-left security** (Snyk, Trivy, Checkov); introduced ephemeral preview environments per PR, cutting QA cycle times significantly.
- Managed EKS with HPA, KEDA, Istio, and External Secrets Operator; drove SOC2, ISO 27001, and GDPR compliance; designed DR strategy with automated AWS Backups.
- Built end-to-end observability and distributed tracing using Prometheus, Grafana, Loki, ELK Stack, Datadog, Sentry, and Grafana Tempo.

IDenTV

DevOps Engineer

Remote

Feb 2020 – Jul 2021

- Deployed ML models on Kubernetes and bare-metal for real-time video analytics; engineered **GPU sharing across pods** using the NVIDIA device plugin to maximise inference utilisation.
- Cut monthly AWS spend from **\$24K to \$12K (50%)** through right-sizing and OpenStack consolidation; provisioned EKS, ECS, RDS, and ElastiCache via Terraform, reducing manual setup time by **70%**.
- Packaged golden AMIs with pre-baked AI models (Packer + Ansible) for **single-command deployments in air-gapped** government environments.
- Established DevOps practice from scratch — CI/CD (CircleCI, TravisCI), Helm charts for ML deployment, HashiCorp Vault for secrets, Flussonic for video streaming, Milvus for vector storage.

Multiple Clients

Freelance DevOps Consultant

Remote

Jun 2021 – Present (Part-time alongside full-time roles)

- **Blockchain:** Deployed HA EVM node/validator infrastructure on EKS with sub-second RPC latency; automated smart contract pipelines with Hardhat and Foundry.
- **Marketplace & Data:** Architected nested CloudFormation templates for AWS SaaS Marketplace; enabled cross-cloud execution across AWS and Azure using Snowflake Container Services; built real-time data pipelines with AWS Glue + Kinesis Firehose via Terragrunt; deployed Grafana Alloy for unified telemetry across client environments.
- **Platform:** Deployed AWS Account Factory for Terraform (AFT) for multi-account vending; designed secure multi-account AWS architectures for crypto analytics clients (Altada Technology Solutions).
- **Data & ML:** Engineered scalable ML inference hosting on EKS/ECS with Milvus for vector storage; conducted chaos testing with LitmusChaos; implemented Cilium for eBPF-based networking.

Early Career

NepalLink Network

Linux System Support Engineer

Kathmandu, Nepal

Apr 2019 – Apr 2020

Managed VMWare VPS environments (LEMP/LAMP), hardened server security (Maldet, ClamAV, CSF), and deployed Icinga monitoring while providing SLA-driven client support.

Vianet Communications Pvt. Ltd.

Junior Network Monitoring Engineer

Kathmandu, Nepal

Apr 2018 – Apr 2019

Automated OLT reboots via Bash; deployed Nagios, Zabbix, Cacti, and CheckMK for proactive fault detection.

Datalytics Pvt. Ltd.

Technical Researcher

Kathmandu, Nepal

Dec 2016 – Feb 2018

Shell scripting for access auditing, Voice Recognition System projects, Cisco network troubleshooting.

Education

Kathmandu University

Bachelor of Engineering (Electrical & Electronics)

Kathmandu, Nepal

2012 – 2016

Tri-Chandra Multiple Campus

Master of Arts in Psychology

Kathmandu, Nepal

2023 – Present

Pursuing alongside career to deepen understanding of team dynamics and organisational behaviour.

Certification



AWS Certified DevOps Engineer – Professional